

Homework Exercises 7

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After completing all lab exercises below, zip all your solution files into a single archive with your student ID (e.g. u210023.zip) and upload it to eClass before the official deadline.

1 Discovering Algorithms

You should apply strategies learned in lecture to solve following puzzles.

1. **Maximum Sum Descent.** Some positive integers are arranged in a triangle like the one shown in Figure 1. Design an algorithm (more efficient than an exhaustive search, of course) to find the largest sum in a descent from its apex to the base through a sequence of adjacent numbers, one number per each level. Express your solution generally using pseudocode. Save your pseudocode in `ps1-1.txt` file.

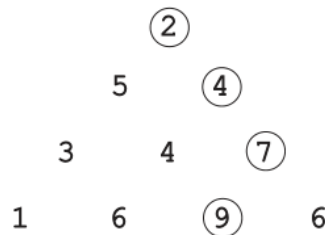


Figure 1: Triangle of integers with the maximum-sum path shown by the circles

2. **Counting Triangles.** An algorithm starts with a single equilateral triangle and on each subsequent iteration adds new triangles all around the outside. The results for the first three values of n are shown in Figure 2. How many small triangles will be there after the n th iteration? Express your solution in formula and save it in `ps1-2.txt` file.

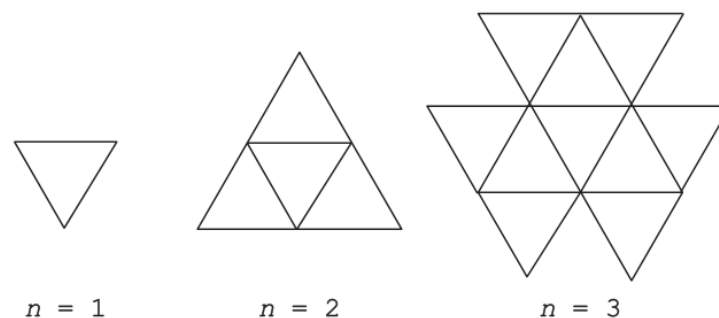


Figure 2: First 3 iterations of Counting Triangles puzzle

3. **Pluses and Minuses.** The n consecutive integers from 1 to n are written in a row. Design an algorithm that puts signs "+" and "-" in front of them so that the expression obtained is equal to 0 or, if the task is impossible to do, returns the message "no solution." Your algorithm should be much more efficient than an examination of all possible ways to place the signs. Express your solution generally using pseudocode. Save your pseudocode in `ps1-3.txt` file.
4. **Toads and Frogs.** On a one-dimensional board with $2n + 1$ cells, there are n counters in the first n cells representing Toads and n counters in the last n cells representing Frogs. Toads and Frogs take turns moving. Moves consist of sliding a Toad or Frog into the empty cell or jumping over one opposing creature to the empty cell. (Toads cannot jump over themselves and neither can Frogs.) Toads can only move rightward; Frogs can only move leftward. The object is to make them switch their positions. For example, for $n = 3$, the task can be depicted by the following diagram:

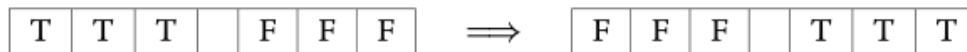


Figure 3: Frogs and Toads

Express your solution generally using pseudocode. Save your pseudocode in `ps1-4.txt` file.

5. **Six Knights.** There are six knights on a 3×4 chessboard: the three white knights are at the bottom row, and the three black knights are at the top row. Exchange the knights to get the position shown on the right of Figure in the minimum number of knight moves, not allowing more than one knight on a square at anytime. Save your solution in `ps1-5.txt` file.

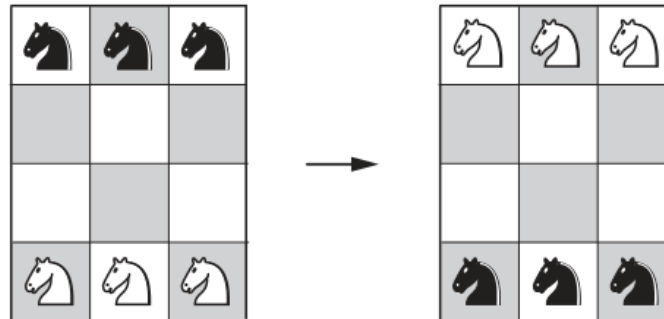


Figure 4: Six Knights

6. **Squid Game.** In a horror game show, there are n people numbered 1 to n standing in a circle. Starting the count with person number 1, every second person is shot until only one person is left. Where in the circle should a person stand to remain the last person standing? Write your answer in `ps1-6.txt` file.

2 Python Tasks

Review **Week 5: Mid-Term Exercises** on this course. Then complete following Python tasks.

1. Write a Python function that returns the sum of the pairwise products of `listA` and `listB`. You should assume that `listA` and `listB` have the same length and are two lists of integer numbers. For example, if `listA = [1, 2, 3]` and `listB = [4, 5, 6]`, the dot product is $1 * 4 + 2 * 5 + 3 * 6$, meaning your function should return: 32

Hint: You will need to traverse both lists in parallel. This function takes in two lists of numbers and returns a number.

Save your program as `ps2-1.py`.

2. Write a Python function that mutates `L` such that it reverses its elements and also reverses the order of the `int` elements in every element of `L`. It does not return anything.

For example, if `L = [[1, 2], [3, 4], [5, 6, 7]]` then `deep_reverse(L)` mutates `L` to be `[[7, 6, 5], [4, 3], [2, 1]]`

Save your program as `ps2-2.py`.

3. Write a Python function that takes list of integers `L` and two functions `f(i)` and `g(i)` as arguments. `f` takes in an integer, applies a function, returns another integer `g` takes in an integer, applies a Boolean function, returns either `True` or `False` Function mutates `L` such that, for each element `i` originally in `L`, `L` contains `i` if `g(f(i))` returns `True`, and no other elements. Function should return the largest element in the mutated `L` or `-1` if the list is empty.

For example, the following functions, `f(i)`, `g(i)`, and test code:

```
def f(i):
    return i + 2
def g(i):
    return i > 5

L = [0, -10, 5, 6, -4]
print(applyF_filterG(L, f, g))
print(L)
```

Should print

```
6
[5, 6]
```

Save your program as `ps2-3.py`.

4. Write a function to flatten a list. The list contains other lists, strings, or ints. For example, `[[1, 'a', ['cat']], 2], [[[3]], 'dog'], 4, 5]` is flattened into `[1, 'a', 'cat', 2, 3, 'dog', 4, 5]` (order matters). Save your program as `ps2-4.py`.